Student GPA Predictor Project Report

Section: BCS-4D

Muhammad Ali Raza (24K-0794)
Muhammad Sarim Hussain (24K-0861)
Aakif Abrar Awan (24K-0889)

1. Introduction

Student academic performance depends on many factors such as study habits, attendance, previous educational background, sleep routine, extracurricular participation, and learning engagement. Educational institutions often analyze these factors to understand student success and identify areas where improvement is needed.

In this project, a machine learning model was developed to predict a student's **College GPA** using academic and lifestyle-related features. The project demonstrates how data analysis and machine learning techniques can be applied in the education sector to estimate student performance.

2. Objectives

The main objectives of this project were:

- To clean and preprocess the academic dataset
- To analyze relationships between student factors and GPA
- To build a Linear Regression model for GPA prediction
- To evaluate model performance using cross-validation
- To create a user input system for predicting GPA

3. Dataset Description

The dataset contains student academic records where each row represents one student.

Target Variable:

- College GPA

Input Features:

- Study Hours per Week
- Attendance Rate
- Major
- High School GPA
- Extracurricular Activities
- Part-Time Job
- Library Usage per Week
- Online Coursework Engagement
- Sleep Hours per Night

These features are used to predict the final College GPA.

4. Data Preprocessing

Before training the model, the dataset was cleaned and prepared.

Missing values were handled by replacing numerical values with the mean and categorical values with the mode. Duplicate rows were removed to ensure data quality.

Categorical variables such as **Major** were converted into numerical form using One-Hot Encoding, while **Part-Time Job** was converted using binary encoding.

Finally, **StandardScaler** was applied to normalize all features so that they operate on a similar scale, improving model performance.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a crucial step in any Artificial Intelligence project, as it helps in understanding the underlying structure, patterns, and relationships within the dataset before applying machine learning models. In this project, EDA was performed to analyze how different student-related factors influence College GPA. Various visualization techniques such as heatmaps and box plots were used to identify correlations, detect outliers, and gain meaningful insights from the data. This step ensures better decision-making during data preprocessing and model building.

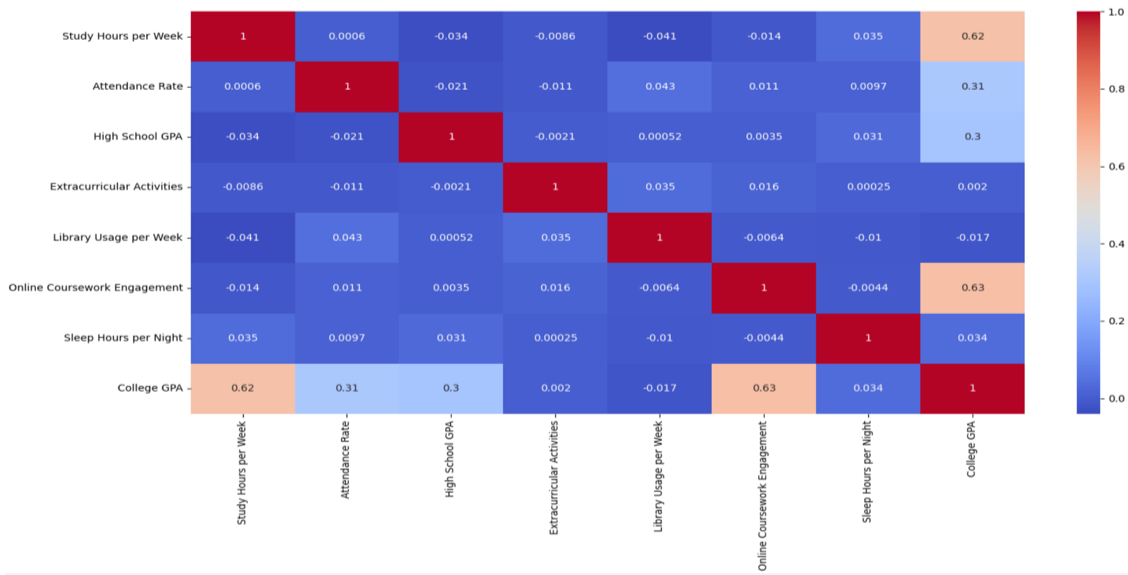
Heatmap Analysis

A correlation heatmap was used to analyze relationships between variables. It showed how strongly each feature is related to College GPA.

It was observed that:

- Study Hours per Week
- Attendance Rate
- High School GPA

have a positive correlation with College GPA. This means students with better study habits and academic background tend to achieve higher GPA.

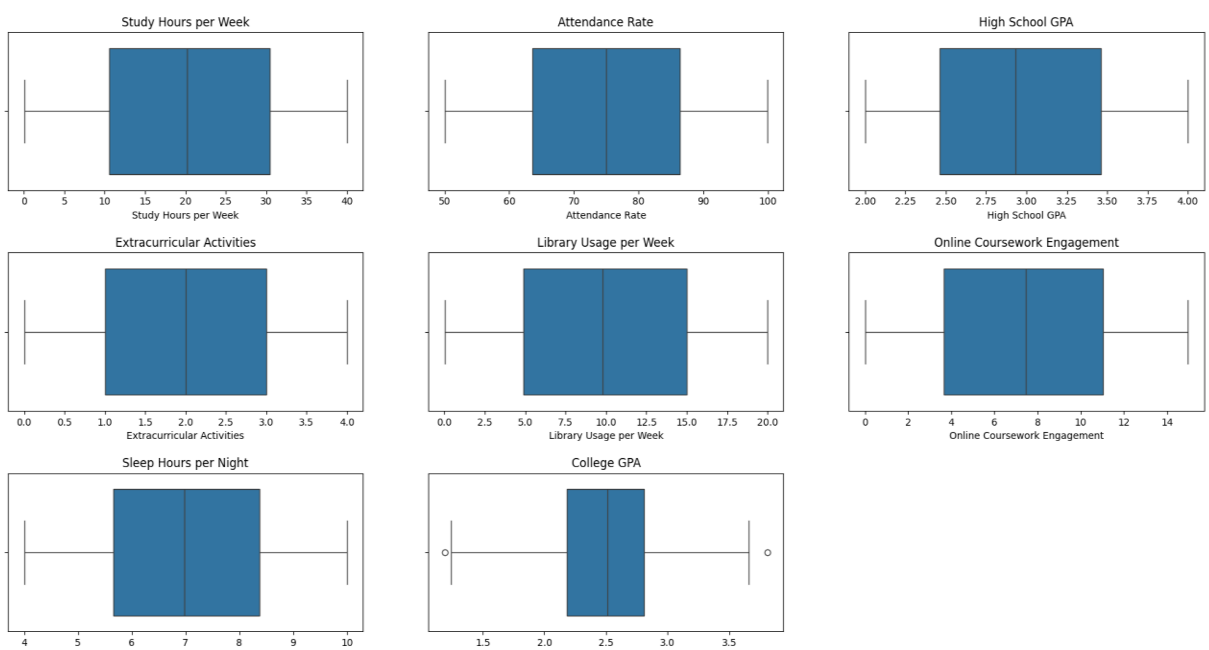


Box Plot Analysis

Box plots were used to analyze the distribution of numerical data and detect outliers.

Some features showed extreme values such as unusually high study hours or low sleep hours. These outliers can affect model performance if not handled properly.

Box plots helped in understanding data spread and identifying abnormal values before training the model.



6. Model Selection

Linear Regression was selected because the target variable (College GPA) is continuous.

The model predicts output using the following equation:

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n$

Where:

- y = predicted GPA
- x = input features
- β = model coefficients

Linear Regression is simple, interpretable, and suitable for regression problems like GPA prediction.

7. Model Evaluation

To evaluate the model, **5-Fold Cross Validation** was used.

The dataset was divided into five parts. In each iteration, four parts were used for training and one part for testing. This process was repeated five times.

The model was evaluated using:

- R^2 Score → measures how well the model explains variance
- Mean Absolute Error (MAE) → average prediction error
- Root Mean Squared Error (RMSE) → penalizes large errors

This method provides a more reliable evaluation compared to a single train-test split.

8. Prediction System

A user input system was developed where students enter:

- Study hours
- Attendance
- Major
- High school GPA
- Extracurricular activities
- Part-time job status
- Library usage
- Online coursework engagement
- Sleep hours

The trained model then predicts the expected College GPA.

9. Results

- Mean R^2 Score = 0.9849197709305091
- Mean Absolute Error = 0.019156056108741954
- Root Mean Squared Error = 0.05476875929286886

These results show how accurately the model predicts student performance.

10. Conclusion

The project successfully developed a machine learning model to predict College GPA using student academic and lifestyle factors.

The results show that study habits, attendance, and prior academic performance significantly influence student outcomes. The project demonstrates the importance of data preprocessing, visualization, and model evaluation in machine learning.

11. Future Improvements

- Use larger real-world datasets
- Apply advanced models like Random Forest or XGBoost
- Improve feature engineering
- Build a GUI or web application
- Add real-time analytics features

12. Tools and Libraries Used

- pandas
- NumPy
- matplotlib
- seaborn
- scikit-learn